



Formulation and characterization of natural deep eutectic solvents (NADES) for simultaneous phenolics and carotenes extraction from fresh oil palm leaf

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ABSTRACT

In this study, natural deep eutectic solvents (NADES), comprising choline chloride (ChCl) as hydrogen bond acceptor (HBA) with hydrogen bond donor (HBD) including glucose (Glu), fructose (Fru), xylose (Xyl), xylitol (Xyli), and glycerol (Gly), were formulated and characterized for simultaneous extraction of phenolic compounds (polar) and carotenoids (non-polar) from fresh oil palm leaves (OPL). pH, density, polarity, Fourier Transform Infrared Spectroscopy (FT-IR) and rheological analyses were conducted, with relation to the extraction efficiency achieved via homogenizer-assisted heating and stirring extraction approach. Among the NADES formulations tested, ChCl:Xyl (1:1) displayed the highest total phenolic content (TPC) at 20.23825 mg/g OPL, due to its optimal polarity at 49.4654 kcal mol⁻¹ and a viscosity of 23288 mPa.s. The highest total carotenoid content (TCC) was observed in ChCl:Xyli (1:1) at 325.94 µg/g OPL, with a lower polarity at 50.0280 kcal mol⁻¹. Despite having similar viscosity and polarity, ChCl:Fru (1:2) with a pH of 5.82 outperformed ChCl:Glu (1:1) with a pH of 6.72 due to enhanced phenolic compound solubility and stability. This study shed light on the potential of NADES, particularly in overcoming the challenge of high moisture content in OPL and other fresh biomasses, which often acted as a barrier during phytonutrients extraction.

1. Introduction

Malaysia, a global palm oil powerhouse, claimed the second spot in 2023 by producing a staggering 18.55 million metric tons of palm oil (Summary Of The Malaysian Palm Oil Industry, 2023). Each oil palm bunch bears between 1000 and 3000 fruits, but only constitutes 10 % of the entire oil palm tree, leaving the remaining 90 % as underutilized waste. What often goes unnoticed, however, is that within this thriving industry, OPL remain an overlooked treasure trove. Despite this substantial resource, OPL has largely escaped the attention of investors and industries. The relative lack of engagement with OPL can be attributed,

in part, to its limited involvement in the palm oil processing chain compared to other biomass components, such as fresh fruit bunches and empty fruit bunches. Consequently, commercialized OPL products remain scarce. Traditionally, OPL has found use in rural communities for crafting rattan baskets (Coomes, 2004), and its fibers have shown potential as papermaking material (Daud and Law, 2011). In a notable breakthrough, OPL was transformed into OPL herbal tea in 2016 (Oil Palm Leaf Green, 2016). According to Ooi et al. (2017), per hectare of palm plantation, approximately 10 tonnes of OPL waste is generated, comprising over 50 % of oil palm biomass. This abundant but neglected biomass becomes a significant environmental concern when left

Abbreviations: ChCl, Choline chloride; DES, Deep eutectic solvent; FC, Folin-Ciocalteu; FT-IR, Fourier Transform Infrared Spectroscopy; Fru, Fructose; Gal, Galactose; Glu, Glucose; Gly, Glycerol; HBA, Hydrogen bond acceptor; HBD, Hydrogen bond donor; OH, Hydroxyl; NADES, Natural Deep Eutectic Solvent; OPL, Oil palm leaf; OPLE, Oil palm leaf extract; TCC, Total carotene content; TPC, Total phenolic content; Xyli, Xylitol; Xyl, Xylose.

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unaddressed, contributing to issues such as eutrophication, haze, and pest infestations within plantations.

The underutilization of OPL results in environmental challenges, however, by extracting phytonutrients from OPL, not only can these environmental burdens be alleviated, but also economic opportunities can be created by adding value to the palm oil industry. Oil palm leaf extract (OPLE) is a rich repository of phytonutrients, including carotenoids, tocopherols, sterols, phenolic compounds, squalene, and coenzyme Q10. These bioactive compounds offer numerous health benefits to humans and are valuable natural resources for various food applications. OPLE is a natural treasure trove of phytonutrients that holds significant promise for enhancing health and well-being. What sets OPLE apart is its impressive content of natural antioxidants, surpassing green tea extract by a substantial 8 % (Mohamed, 2014). This heightened antioxidant potency is primarily attributed to the abundance of non-toxic phenolic compounds within OPLE. The comprehensive array of phytonutrients present in OPLE is a testament to its potential in promoting health. Delving into its diverse health effects, OPLE shines as a formidable agent against cancer and tumours. When administered to rodents, it effectively prevented and treated breast tumours (Namvar et al., 2012). Beyond this, OPLE plays a crucial role in supporting the regeneration and repair of organs and tissues, with a particular focus on the liver. In the context of liver health, OPLE acts as a protective shield by reducing blood pressure, enhancing vascular function, mitigating liver inflammation, and shielding against harm inflicted by toxic aldehydes. Additionally, OPLE showcases antibacterial and anti-inflammatory properties, which render it a valuable contender for modulating the immune system. Its influence on the immune system's response to injuries and swelling is notable (Prasain et al., 2010). In the realm of chronic health conditions, OPLE shows promise. Research indicates its potential in modulating the immune system, crucial for ailments like aging-related issues, cancer, diabetes, heart problems, allergies, and AIDS. Studies on older rodents revealed OPLE's dual-phase impact on immune response: at lower doses, it initially increased swelling before reduced it, while at higher doses, it significantly decreased swelling (Prasain et al., 2010). In summary, OPLE, with its rich repository of phytonutrients and diverse health benefits, emerges as a natural powerhouse for promoting overall health. From its antioxidant prowess to its anti-cancer effects, organ protection, immune system modulation, and its role in addressing chronic health conditions, OPLE stands as a testament to the abundant potential of nature's offerings in supporting our well-being.

Historically, phytonutrient extraction from OPL has relied on conventional organic solvents such as hexane, benzene, methanol and chlorocarbons (Easmin et al., 2015). However, it is not the best option for extraction because it is corrosive, non-biodegradable, vaporous, toxic and are harmful to human. Even though supercritical fluid is non-toxic, inert, abundant and readily removable, the production of supercritical fluid requires high pressure equipment which is potentially costly and possesses higher risk. On top of that, supercritical fluid has high polarity which makes them less efficient in extracting non-polar phytonutrients. Ionic liquid (IL) has gained much attention in the past decade due to their flexibility. Their physical, chemical and biological characteristics can be tailored through combining different cation and anion to meet the desired characteristics. However, their biodegradability, biocompatibility and toxicity are frequently questionable. Although water is considered as a universal green solvent, their efficiency in extracting organic compounds are usually low.

To counter the challenges associated with supercritical fluids, IL, water, and the conventional extraction media, a relatively economical and environmentally friendly solvent, known as NADES, can be used. NADES represents a novel solvent class originating from deep eutectic solvents (DES). NADES comprise DES combined with naturally abundant, biodegradable, non-toxic, non-volatile, and inexpensive components, resulting in a favorable environmentally friendly profile. First introduced in 2011, NADES aimed to address the solubility of intracellular compounds insoluble in both water and lipid phases. It was

identified as a third phase within living organisms, facilitating the transport of hydrophilic and lipophilic compounds, and participating in the biosynthesis, solubilization, and storage of various hydrophobic metabolites and unstable compounds in cells. In recent years, NADES have presenting themselves as efficient alternatives to conventional IL with enhanced cost-effectiveness. Their straightforward synthesis and the adaptability in selecting constituent components render them more accessible than intricate and expensive IL and DES. This has catalysed their incorporation into various applications within the realm of food processing. NADES share many of the commendable attributes of IL, encompassing biodegradable constituents, non-flammability attributable to their minimal or indiscernible vapour pressure, and low toxicity.

NADES is a promising green organic solvent that is formed through the complexation of HBA and HBD using natural products such as primary metabolites (Nystedt et al., 2021; Yang, 2019). HBA is generally a salt anion such as ammonium salt or amino acid, whereas HBD is mainly sugar, organic acid or alcohol. The common HBA used in previous research is ChCl , citric acid and urea. Additionally, sugars such as Glu, Xyl, and Fru, along with sugar alcohols like Xyli and Gly, have been utilized in the synthesis of NADES. The use of NADES as extraction solvents appears to be a potential strategy in several industries, including nutraceuticals, pharmaceuticals, food, cosmetics, and green chemistry. This potential arises from the unique physicochemical properties, cost-effectiveness, low toxicity, biodegradability, low vapor pressure, minimal flammability, thermally stable and their biological activity of NADES. Studies have reported that NADES exhibit high extraction efficiency on different biomass at low toxicity. Therefore, they represent an appealing greener alternative compared to predecessor IL and traditional organic solvents for extracting and stabilizing both polar and non-polar phytonutrients from natural waste materials.

The potential of NADES as extraction and preservation media for phytonutrients has been discussed in numerous works of literature as depicted in Table S1 which shows the credibility of NADES and effectiveness of NADES in these applications. Extraction of bioactive compounds such as phenolic, flavonoid and anthocyanin using NADES has been tested and showed great yield as compared to other conventional organic solvent such as methanol and ethanol. Methanol is a conventional organic solvent commonly used in extractions. However, several reviews have highlighted that NADES offer higher extractability compared to methanol. For example, the solubility of rutin in NADES significantly exceeds that in water or methanol (Huang et al., 2017; Zhao et al., 2015). Specifically, the solubility of rutin in ChCl -based and Gly-based NADES was 660–1577 times higher than in water and 1.46 times higher than in methanol. The extraction of phytonutrients such as phenolics using NADES has been done extensively, for instance, Wei et al. (2015) conducted an extraction of phenolics from *Cajanus cajan* leaves using NADES. It was found that ChCl :Maltose exhibited the highest extractability (30.63 mg/g) with a molar ratio of 1:2. Additionally, Zain et al. (2021) evaluated the extractability of NADES on flavonoids, luteolin, and apigenin from OPL. The results indicated that NADES demonstrated moderate extractability for flavonoids (12.67 mg/g to 13.92 mg/g quercetin equivalents (QCE)) and apigenin (117.61 mg/g to 128.18 mg/g), while showing high efficiency in extracting luteolin (16.12 mg/g to 18.98 mg/g orientin equivalents (OE)). Furthermore, NADES possess a distinctive property of enhancing the solubility of a broad spectrum of poorly water-soluble substances. For instance, the solubility of rutin in ChCl -based NADES (ChCl :Sucrose, ChCl :Glu, ChCl :Fru, ChCl :Malic acid, ChCl :aconitic acid) was found to be 50–100 times higher than in water, ranging from 20 g/kg to 80 g/kg (Choi et al., 2011). The researchers also demonstrated that NADES consisting of ChCl and Glu, could dissolve water-insoluble compounds such as starch, paclitaxel, and ginkgolide B. On top of that, NADES also bolster the stability of bioactive elements owing to their robust hydrogen bonding affinity with the targeted compounds (Li, 2022). Phytonutrients are prone to degradation or changes under specific conditions like high temperatures, pH variations, or enzymatic

reactions. These constraints impede the widespread utilization of phytonutrients. Nevertheless, NADES provide a remedy to these issues. In a study by Jeliński et al. (2019), the stability of curcumin was assessed in a conventional extraction solvent namely methanol and NADES (ChCl: Gly) under artificial sunlight exposure for 120 minutes. While the concentration of curcumin in methanol dwindled to 5 % of its initial level, curcumin remained unchanged in NADES throughout the experiment, manifesting no signs of degradation. Similarly, the stability of baicalin, curcumin, andrographolide, and oleanolic acid was examined in a 0.9 % NaCl solution and NADES formulated with food and pharmaceutical additives (ChCl, betaine, acetamide, organic acid, and alcohol) (Cao et al., 2020). All the bioactive compounds were effectively conserved in NADES even when subjected to high temperatures at 80 °C for 6 hours, displaying superior stability compared to the saline solution.

The extraction of bioactive compounds from OPL mainly focused on phenolics and flavonoids only, as refer to Table S1. There is no study on the simultaneous extraction of phenolics and carotenes from fresh OPL using NADES, thus, the extractability of these phytonutrients simultaneously through the use of NADES has not been studied before. Hence, interest remained, and studies are required on the breadth and depth to reveal the full potential of NADES on the extractability of phytonutrients from fresh OPL which can be further instigated for other applications. Based on the current available study, the extractions of bioactive compounds from OPL only focused on dried OPL and there is no study on fresh OPL. The use of fresh OPL excluded lengthy drying and milling, thus saving time, cost and energy. However, fresh OPL contained high moisture content which acted as a barrier between the targeted nutrient and organic solvent, thus reduced the extractability of organic solvents. In this work, the problem is solved by the use of polarity-tunable and water-soluble NADES, which was able to penetrate the wet material freely for the extraction of phytonutrient. The TPC and TCC yields were analysed with the support of the polarity, pH, and viscosity of NADES. The measurement of TCC in OPL NADES extract was a challenge and almost impossible via absorption spectroscopy due to the interference of chlorophyll. In this paper, a simple and efficient procedure is presented for the first time after solid phase extraction and ion exchange.

This study focuses on formulating and characterizing stable NADES for extracting phytonutrients from OPL, while also presenting simple analytical methods for measuring phytonutrients in NADES extracts. By creating these analytical protocols, the process of measuring phytonutrients in NADES extracts was simplified, offering valuable tools for accurately assessing the phytonutrient content of the extracts. Additionally, the extractability of phenolic and carotenoid compounds in the formulated NADES was assessed. Through rigorous experimentation and analysis, we aim to shed light on the efficacy of NADES in extracting these bioactive constituents from OPL, providing critical insights into their performance and versatility as extraction solvents for a diverse range of phytonutrients.

2. Materials and methods

2.1. Chemicals

OPL (*Elaeis guineensis* leaves) was collected from Malaysian Palm Oil Board (MPOB), Bangi and stored in the freezer after being washed and dried. D-(+)-Glucose anhydrous (CAS No. 50–99–7; Chemiz; >99.5 %), D-(-)-Fructose (CAS No. 57–48–7; Chemiz; >99.5 %), D-(+)-Xylose (CAS No. 58–86–6; Sigma-Aldrich; ≥ 99.0 %), D-galactose (CAS No. 59–23–4; EMD Millipore Corp; ≥ 97.0 %), choline chloride (CAS No. 67–48–1; Sigma-Aldrich; ≥98.0 %), Xylitol (CAS No. 87–99–0; Sigma-Aldrich; ≥99.0 %), Glycerol (CAS No. 56–81–5; Chemiz; >99.9 %), gallic acid anhydrous (CAS No. 149–91–7; Chem Soln; ≥99.0 %), sodium carbonate anhydrous, Na₂CO₃ (CAS No. 497–19–8; Chem Soln; ≥99.8 %), Folin-Ciocalteu's phenol reagent (CAS No. 7791–20–0; Chem Soln), acetone (CAS No. 67–64–1; Chemiz; ≥99.9 %), methanol (CAS No. 67–56–1; Elite Advanced Materials Sdn. Bhd.; ≥ 99.9 %), hexane (CAS

No. 100–54–3; Chem Soln; >95.0 %); β-carotene (CAS No. 7235–40–7; Sigma-Aldrich; ≥93.0 %), Nile Red (CAS No. 7385–67–3; Thermo Scientific; 99.0 %), and Ambersep 900 hydroxide (CAS No. 9017–79–2; Sigma-Aldrich) were acquired for the experiments.

2.2. Preparation of NADES

The NADES employed in this study is outlined in Table 1. ChCl was selected as the HBA for the synthesis of NADES, a choice substantiated by its favourable attributes, including its classification as a vital nutrient, cost-effectiveness, non-toxicity, biodegradability, and historical relevance as a precursor to vitamin B4. In tandem, Glu, Xyl, Gal, Gly and Xyli functioned as the HBD to facilitate the formation of eutectic mixtures with ChCl. Consequently, distinct molar ratios—1:1, 1:2, 2:1 and 1:4—were systematically synthesized to investigate the compatibility and synergistic interplay between HBA and HBD in the formulation of NADES. The NADES synthesis process involved the mixing of HBA and HBD constituents at precise molar ratios within securely sealed screw-cap vials. Subsequently, this mixture was subjected to a controlled process characterized by continuous heating and stirring at a constant temperature of 80 °C in an oil bath, facilitated by a magnetic hot plate stirrer. Heating and stirring was persisted until a clear homogenous liquid was formed. The homogeneous clear solution was allowed to undergo a gradual and natural cooling process, ultimately reaching ambient room temperature. The synthesized NADES were assessed for their physical characteristics and stability, focusing on attributes such as uniformity, colour, phase separation, agglomeration, and the formation of moulds. These observations were made after the NADES had been allowed to cool down to room temperature and were recorded every 24 hours for 7 days.

2.3. pH of NADES

The pH measurement for NADES samples was conducted at room temperature using a calibrated portable pH meter, specifically the Testo 206 – pH2 Starter Set. This pH meter is well-suited for determining pH values in semi-solid materials, viscous substances, and food products containing proteins. In the measurement process, the pH bulb of the pH meter was fully immersed in the NADES, and the measurement button was activated. The pH meter was kept in the solution until the reading had stabilized, at which point the pH value was recorded. Subsequently, the electrode was rinsed with distilled water and thoroughly dried before being employed for pH measurement in other NADES samples. The pH analysis was carried out in triplicate, and the average pH value was determined from the obtained readings.

2.4. Density of NADES

The density of NADES was measure gravimetrically via 5 mL measuring cylinder. The mass (M_{NADES}) and volume (V_{NADES}) were

Table 1
NADES formulations employed in this study.

NADES		Molar ratio	Sample name
HBA	HBD		
Choline chloride	Glucose	1:1	ChCl:Glu (1:1)
	Glucose	1:2	ChCl:Glu (1:2)
	Glucose	2:1	ChCl:Glu (2:1)
	Xylose	1:1	ChCl:Xyl (1:1)
	Xylose	1:2	ChCl:Xyl (1:2)
	Xylose	2:1	ChCl:Xyl (2:1)
	Galactose	1:1	ChCl:Gal (1:1)
	Galactose	1:2	ChCl:Gal (1:2)
	Galactose	2:1	ChCl:Gal (2:1)
	Glycerol	1:4	ChCl:Gly (1:4)
	Xylitol	1:1	ChCl:Xyli (1:1)

carefully measure for density (ρ_{NADES}) calculation through Eq. (1).

$$\rho_{\text{NADES}} = \frac{M_{\text{NADES}}}{V_{\text{NADES}}} \quad (1)$$

2.5. Polarity of NADES

Polarity measurements were carried out using Nile red as solvatochromic probe. The stock solution was prepared by dissolving 1 g of Nile Red in 1 L of ethanol and stored in the refrigerator at 4 °C, maintaining its integrity until deployment. The NADES sample was placed in a 1 cm² quartz cuvette, and a blank was recorded. Precisely 60 µL of the Nile Red stock solution was pipetted and then introduced into the quartz cuvette containing the NADES. A mixing process ensued to ensure a homogeneous amalgamation of the Nile Red stock solution with the NADES matrix. To eliminate any residual ethanol, the sample was subjected to a controlled high-purity nitrogen gas stream. Mass measurements were repeatedly conducted until a stable sample mass was consistently obtained, providing assurance of complete ethanol evaporation. Subsequently, the sample's absorbance was measured within the wavelength spectrum spanning from 400 to 700 nm, employing a UV-Vis spectrophotometer. The wavelength exhibiting the peak absorbance, denoted as λ_{max} , was precisely identified. The calculation of the molar transition energy, E_{NR} was obtained by employing Eq. (2) (Dai et al., 2015; Zain et al., 2021), where E_{NR} is molar transition energies in kcal mol⁻¹, λ_{max} is the maximum wavelength in nm, h is Planck's constant at 6.626×10^{-34} kg.m².s⁻¹, c is speed of light in vacuum at 3×10^8 ms⁻¹ and N_A is Avogadro's constant at 6.022×10^{23} particles mol⁻¹.

$$E_{\text{NR}} = hcN_A/\lambda_{\text{max}} \quad (2)$$

2.6. Fourier Transform Infrared Spectroscopy (FT-IR) analysis

FT-IR analysis was used to detect the functional groups of the NADES. The spectra were recorded using FT-IR spectroscopy (FT-IR Perkin Elmer, Attenuated total reflection, ATR) with the transmission spectra measured in the range of 4000 cm⁻¹ to 400 cm⁻¹ at a resolution of 4 cm⁻¹.

2.7. Rheology of NADES

The rheology test of the formulated NADES was measured at room temperature using a rheometer (Anton Paar MCR 72) with a stainless-steel parallel plate geometry (PP25, diameter 25 mm, gap 1 mm) for soft solid or gel-like samples. The measuring position (gap) and load position (height) were initialized before analysis, and the moving profile was set as 'viscoelastic'. Rheology test with a shear rate from 0.1 to 100 s⁻¹ and the rheology profile was performed and obtained from Anton Paar RheoCompass software.

2.8. Extraction of TPC and TCC from fresh OPL

OPL underwent an initial cleaning process, followed by the removal of leaf stalks, and then the leaflets were finely cut into small pieces, each measuring approximately 1 mm × 10 mm. The extraction process was conducted utilizing NADES, facilitated by a homogenizer (IKA T10 Ultra Turrax) operating at a power level of 2 for a duration of 2 minutes. The mixture was then transferred into a glass bottle and extracted via heating and stirring. Key extraction parameters included 40 mL/g solid-to-liquid ratio, 10 wt% water content, an extraction temperature of 40°C, and a 40-minute extraction duration. Subsequently, a centrifugation step was employed to separate and obtain the supernatant. This centrifugation process was conducted at 3000 revolutions per minute (rpm) for a duration of 10 min. The quantification of the total extraction yield was determined through UV-Vis spectrophotometry, allowing for quick and precise measurement and analysis of the extracted

components. Extractions for each parameter were duplicated, and the mean TPC and TCC were determined based on the collected values with relative standard deviation (RSD) of less than 15 %.

2.9. Gallic acid standard calibration curve

The standard calibration solution for phenolic compounds was established using gallic acid as the reference. A 1 mg/mL stock solution was prepared by accurately combining 25 mg of gallic acid and dissolving it in 25 mL of ethanol. This 1 mg/mL stock solution was subsequently diluted to concentrations of 0.5, 0.25, 0.125, 0.0625, 0.03125, and 0.015625 mg/mL using ultrapure water. For the analysis, 0.1 mL of each of these stock solutions was carefully transferred to separate test tubes and mixed with 0.5 mL of a 10 % Folin-Ciocalteu (FC) solution, alongside 7 mL of ultrapure water. The FC reagent comprises sodium tungstate and sodium molybdate within an acid medium. During the reaction, the initially yellow-coloured FC reagent interacted with the phenolic compounds, leading to a colour change from yellow to blue as it got reduced by the phenolic compounds. The stock solutions, each with varying concentrations, were stored in darkness for a duration of 10 min. Following this incubation period, 1.5 mL of a 7.5 % sodium carbonate solution was added to each stock solution, and they were once again kept in darkness for 2 hours at room temperature. Subsequently, the absorbance of each stock solution was meticulously recorded at a wavelength of 765 nm, and a calibration curve was constructed for further analytical purposes.

2.10. Carotene standard calibration curve

The calibration standard solution for carotenoid compounds was established utilizing β -carotene. A 15 µg/mL stock solution was formulated by accurately dissolving 750 µg of β -carotene in 50 mL of acetone. Subsequently, this 15 µg/mL stock solution was diluted to various concentrations, specifically 0.1, 0.5, 1.5, 3.0, 6.0, and 9.0 µg/mL, using acetone as the diluent. The absorbance of each of these stock solutions was determined at a wavelength of 450 nm, and a calibration curve was subsequently generated to facilitate further analysis.

2.11. Measurement of TPC in NADES extract

A volume of 0.1 mL of the NADES extract was carefully withdrawn using a micro syringe and then transferred into a screw cap vial. To this vial, 0.5 mL of a 10 % FC solution and 7 mL of ultrapure water were added. The mixture was stirred until complete homogenization was achieved. The sealed vials were then placed in a dark environment and left undisturbed for a duration of 10 minutes. Following this incubation period, 1.5 mL of a 7.5 % sodium carbonate solution was introduced into each vial. The vials were once again stored in a dark setting, this time for a period of 2 hours, all at room temperature. Subsequently, the absorbance of the extract solutions was measured using UV-Vis spectrophotometry at a wavelength of 765 nm. Distilled water served as the blank sample for the determination of TPC. If the absorbance of the extract solution fell outside the range of the standard calibration graph, sample dilution was carried out as necessary. This entire process was repeated twice with RSD less than 15 %, and the average TPC value was calculated for each NADES extract.

2.12. Measurement of TCC in NADES extract

Solid phase extraction was executed using a solid phase cartridge (Bond Elut Plexa cartridge, 200 mg, 3 mL). Prior to analysis, the cartridge underwent a conditioning process. Methanol was first used to flush through the solid phase, followed by distilled water to activate the adsorbent within the cartridge. Given the high viscosity of the NADES extract, 5 mL of ultrapure water was added to the extract solution and vigorously vortexed until achieving homogeneity. The diluted extract

solution was then carefully loaded into the solid phase extraction cartridge, allowing the NADES extract to elute from the cartridge while discarding the outflow. Subsequently, the cartridge was rinsed twice with acetone to extract the carotenoids. To address the presence of chlorophyll in the sample solution, Ambersep 900OH resin was employed for its removal, adhering to a resin-to-sample ratio of 1 g resin to 10 mL of the sample solution. The sample solution with the resin was stirred at room temperature (300 rpm) for 30 minutes, and this procedure was repeated twice with fresh resin. Finally, the absorbance of the sample solution was measured using UV-Vis spectrophotometry at a wavelength of 450 nm, with acetone serving as the blank solution. The TCC for each extract solution from various NADES was measured and the average value was obtained, with RSD less than 15 %.

2.13. Statistical analysis

Results presented in tables and graphs are expressed as mean ± standard deviation (SD). All experiments were repeated in duplicate, and the values were averaged. The mean values were considered significantly different when $p < 0.05$.

3. Results and discussion

3.1. NADES formation

ChCl, Glu, Xyl, Gal, Fru, Gly, and Xyli were employed to synthesize NADES at varying molar ratios, as documented in Table 2. The HBA and HBD components underwent a process of heating and stirring at 80°C until clear and homogenous liquids were obtained. The heating step was crucial for NADES formation, as it facilitated the dissolution of solids by reducing their cohesive forces and promoted the collision between ChCl and the HBD towards hydrogen bonds formation, thus suppressing the melting point, making the NADES present as a liquid (Jurić et al., 2021; Sekharan et al., 2021). Upon heating, the NADES mixtures exhibited white translucent viscous gel appearance in the initial stages. Subsequently, as the heating and stirring progressed for 120–180 minutes, clear and homogenous liquids were achieved for NADES composed of ChCl:Glu, ChCl:Xyl, ChCl:Gly and ChCl:Xyli. These observations aligned

with previous studies on the formulation of NADES (Biernacki et al., 2020; Gao et al., 2023; Hayyan et al., 2012, 2013).

Both ChCl:Fru (1:1) and (1:2) showed pale-yellow colour after being cooled down to room temperature. This was due to the degradation of Fru during the formation of NADES, causing pale yellow colour for the Fru-based NADES. Extended heating at higher temperatures led to sugar molecule oxidation, causing the liquid to alter from transparent to dark brown due to caramelization (Jurić et al., 2021). Fru is a ketose sugar, while Glu and Xyl are aldose sugars. Ketoses generally exert higher reactivity compared to aldoses due to the presence of keto groups ($-C=O$) instead of an aldehyde groups ($-CHO$). Hence, Fru was prone to undergo caramelization reactions (Istasse and Richel, 2020). Moreover, Fru has a lower melting point than Glu. Notably, in this group of Glu, Gal, Xyl, and Fru, Fru stands out with the lowest melting point at approximately 103–105°C, while Glu has the highest melting point at around 146–150°C, followed by Gal at about 131–134°C, and Xyl at roughly 140–145°C. This lower melting point allowed Fru to heat up quickly and initiated the caramelization process at lower temperatures (Hurtt et al., 2004). To prevent this undesirable outcome, it is essential to carefully control and maintain the heating temperature and heating time.

On the other hand, NADES composed of ChCl and Gal, at all three molar ratios, consistently appeared as white viscous liquids even after continuous heating for 360 minutes. Furthermore, these NADES remained in a white opaque liquid state, as observed for ChCl:Gal at the three tested molar ratios (1:1, 1:2, and 2:1). The NADES which consisted of Gal, separated into two layers after 24 hours at room temperature. The failure of a stable eutectic mixture formation at room temperature was due to its higher eutectic melting point. As a result, the mixture exhibited phase separation or non-uniform distribution, which led to opaque and non-homogeneous appearance.

The establishment of a stable NADES is of paramount importance, as it significantly influences the overall stability of the NADES formulation, thereby expanding its potential for diverse applications. In this context, the attainment of NADES stability plays a critical role in preserving the stability of the phytonutrients extracted from the OPL. By ensuring the stability of the NADES, the extracted phytonutrients are protected in optimal condition over an extended duration, facilitating their utilization in various applications with enhanced efficacy and potency.

ChCl:Glu (1:1), ChCl:Xyl (1:1), ChCl:Fru (1:1), ChCl:Fru (1:2) and ChCl:Gly (1:4) remained as clear liquid after storage at room temperature for 24 hours. However, the Glu-based and Xyl-based NADES at a molar ratio of 1:2 and 2:1 turned translucent due to recrystallization of the eutectic mixture. After 7 days, Glu-based and Xyl-based NADES at a molar ratio of 1:2 and 2:1 exhibited phase separation, resulted in the formation of two distinct phases. The upper phase appeared as colourless and slightly viscous liquid characteristic of the NADES, while the lower phase manifested as white solid, represented the unreacted or undissolved excess sugar and/or ChCl. The occurrence of phase separation was attributed to the high concentration of the HBD or HBA, which led to an imbalance in the halide ions (Cl^-) and hydroxide ions (OH^-) available for hydrogen bonding. As a result, the excess remained unbound and persisted as solid phase. Consequently, the mixture exhibited a division into two phases.

After 7 days of storage, the ChCl:Xyl (1:1) was also found to undergo recrystallization. Xyl, being a pentose sugar, possesses fewer carbon atoms compared to Glu and Fru, which are hexose sugars. It has been documented in Andary et al. (2021) that pentose sugars are generally less stable than hexose sugars. Moreover, chromatographic analysis indicated that the intensity of Xyl degradation was greater than that of Glu. Consequently, Xyl was susceptible to both crystallization and degradation when compared to Glu. The scientific data obtained from the experiments supports the notion that Xyl-based NADES are prone to crystallization and degradation due to the unique characteristics of the pentose sugar Xyl when compared to the more stable hexose sugar Glu. Similar observations have been made with Xyli, which crystallized easily

Table 2
Appearance of NADES.

NADES	Appearance after formulation	Appearance after 7 days
ChCl:Glu (1:1)	Transparent viscous liquid	Transparent viscous liquid
ChCl:Glu (1:2)	Transparent viscous liquid	Solid crystals formed
ChCl:Glu (2:1)	Transparent viscous liquid	Solid crystals formed
ChCl:Xyl (1:1)	Transparent viscous liquid	Translucent liquid with solid crystals
ChCl:Xyl (1:2)	Transparent viscous liquid	Solid crystals formed
ChCl:Xyl (2:1)	Translucent viscous liquid	Solid-liquid separation with solid crystals
ChCl:Gal (1:1)	White opaque viscous liquid	Solid-liquid separation with white opaque gel
ChCl:Gal (1:2)	White opaque viscous liquid	Solid-liquid separation with white opaque gel
ChCl:Gal (2:1)	White opaque viscous liquid	Solid-liquid separation with white opaque gel
ChCl:Fru (1:1)	Clear yellow viscous liquid	Clear yellow viscous liquid
ChCl:Fru (1:2)	Clear yellow viscous liquid	Clear yellow viscous liquid
ChCl:Gly (1:4)	Transparent viscous liquid	Transparent viscous liquid
ChCl:Xyli (1:1)	Transparent viscous liquid	Solid crystals formed

due to its chemical structure in the form of pentose sugar alcohol.

In summary, ChCl:Glu (1:1), ChCl:Fru (1:1), ChCl:Fru (1:2) and ChCl:Gly (1:4) stood out as the preferred options for creating stable NADES. Notably, these NADES maintained their stability even when stored at room temperature for an extended period of 3 months. ChCl:Xyl (1:1) experienced recrystallization after 7 days, indicating decreased stability. The higher susceptibility of Xyl-based NADES to crystallization and degradation compared to Glu-based NADES was attributed to the structural differences between the pentose and hexose sugars. Furthermore, ChCl:Gal NADES consistently separated into two layers, revealing inherent instability at room temperature.

3.2. pH of NADES

The significance of considering the pH of a solvent during the selection process cannot be overstated, as it plays a pivotal role in determining the chemical reactivity and stability of the solution. When working with diverse chemical species and reactions, the pH of the medium becomes an essential factor in preventing unwanted side reactions and ensuring the successful outcome of the intended chemical processes. The pH will also determine the metal selection in various industrial applications during the consideration of corrosion and it also determines the level of toxicity to the living organism.

In the realm of eco-friendly solvent alternatives, NADES have garnered considerable attention for their sustainable properties and potential applications in various fields. However, to meet the criterion of being environmentally friendly, it is imperative that these NADES exhibit a pH ranging from neutral to mildly acidic. This pH range not only contributes to their safety but also facilitates their proper disposal, minimizing any adverse environmental impact (Hayyan et al., 2012, 2013). Interestingly, while eco-friendly NADES typically possess neutral to low acidity for safe handling and disposal, there is another intriguing aspect of NADES with high acidity. These high acidity NADES demonstrate remarkable antimicrobial properties, turning them into compelling candidates for a completely different set of applications. By creating an environment with lower pH levels, these NADES create an inhospitable setting for bacteria and viruses, leading to their elimination (Rachmaniah et al., 2020). Such antimicrobial effects hold great promise in various medical and industrial applications where the control and prevention of microbial growth are of utmost importance.

Table 3 shows the pH of the NADES tested at room temperature. The pH value of the NADES increased from ChCl:Gal (1:2) < ChCl:Gal (2:1) < ChCl:Gal (1:1) < ChCl:Fru (1:1) < ChCl:Fru (1:2) < ChCl:Xyl (2:1) < ChCl:Gly (1:4) < ChCl:Glu (2:1) < ChCl:Xyl (1:2), ChCl:Xyli (1:1) ChCl:Glu (1:2) < ChCl:Glu (1:1) < ChCl:Xyl (1:1). Based on the data, these NADES can be categorized as neutral NADES and slightly acidic NADES.

The NADES comprising ChCl and Gal at varying molar ratios revealed discernible presence of acidity, with pH values ranged from 3.72 to 4.45. This acidity was influenced by the stability characteristics of the Gal-based NADES. Notably, Table 2 elucidates that NADES containing Gal exhibited an incapability to achieve a stable and clear liquid state, instead manifesting as white, opaque, and viscous substances, even under prolonged heating and stirring conditions. The contributory factors to the overall acidity encompassed the presence of free halide ions from ChCl, along with the ChCl molecules. It is important to highlight that ChCl is classified as a weak acid, bearing a pH range typically ranging between 4 and 6 (Registration Dossier, 2004). Furthermore, both Gal and ChCl exemplified hygroscopic properties, rendered them susceptible to moisture absorption from the ambient environment (Brusas and Dela Pena, 2021; van der Sman, 2023). As moisture was absorbed into the NADES mixture, the free halide ions exerted strong attraction to water molecules because of the colloidal behavior of Gal which prevented ChCl from binding to it. As a result of this interaction, hydrochloric acid was formed, which led to a decrease in the pH of the entire mixture. This interplay of factors underscored the intricacy of the acidity observed in the ChCl:Gal NADES, which enriched

Table 3
pH of NADES.

NADES	pH
ChCl:Glu (1:1)	6.72 ± 0.02
ChCl:Glu (1:2)	6.66 ± 0.10
ChCl:Glu (2:1)	6.52 ± 0.13
ChCl:Xyl (1:1)	6.85 ± 0.08
ChCl:Xyl (1:2)	6.66 ± 0.02
ChCl:Xyl (2:1)	6.07 ± 0.41
ChCl:Gal (1:1)	4.45 ± 0.03
ChCl:Gal (1:2)	3.72 ± 0.31
ChCl:Gal (2:1)	4.29 ± 0.12
ChCl:Fru (1:1)	5.73 ± 0.10
ChCl:Fru (1:2)	5.82 ± 0.03
ChCl:Gly (1:4)	6.30 ± 0.50
ChCl:Xyli (1:1)	6.66 ± 0.37
ANOVA (Source of Variation)	
Within group	
SS	2.7903
df	33
MS	0.0846
Between group	
SS	42.3127
df	12
MS	3.5261
F	41.7018
P-value	1.97E–16
F crit	2.0595

The data are expressed as mean ± SD based on duplicate values.

the comprehension of complex behavior inherent in these eutectic systems.

NADES derived from Fru also demonstrated slightly acidic nature, displaying a pH range of 5.73–5.82, which was moderately neutral when contrasted with Gal-based NADES. Both Fru and ChCl induced mildly acidic environment, with pH values spanning from 4 to 6 (Registration Dossier, 2004). The regulated pH of the constituent components contributed to the faintly acidic characteristics of Fru-based NADES. Similar pH result was obtained by Hayyan et al. (2012) and Nystedt et al. (2021) on Fru-based NADES at room temperature. The result aligned with the review done by Skulcova et al. (2019), whereby the author concluded that the Fru based NADES has slightly lower pH as compared to the Glu based NADES.

On the other hand, the NADES containing Gly, Xyl and Xyli exhibited neutral pH with pH range of 6–7 at room temperature. The result was supported by the observation obtained by Hayyan et al. (2013) whereby the author mentioned that the pH of the Glu-based NADES ranged from 6.03 to 7.11 at room temperature. The pH of the eutectic mixture was very much dependant on the type of HBD used in the formation of the NADES. The HBD used in the formation of NADES were Gly, Xyli, Glu and Xyl which exhibited neutral pH. It is also worth mentioning that among the three molar ratios tested in Glu and Xyl based NADES, the NADES composed of Xyl and Glu in a 2:1 ratio showed the lowest pH value. This acidity was attributed to the presence of excess ChCl in the eutectic mixture. As mentioned earlier, ChCl is classified as a weak acid, with a pH range typically between 4 and 6. The higher concentration of ChCl in the 2:1 molar ratio NADES contributed to the observed decrease in pH, making it the most acidic among the tested ratios. Of all the tested NADES, those comprising Glu, Xyl, Gly, and Xyli emerged as particularly promising candidates, distinguished by their neutral pH.

The ANOVA results for the pH of NADES formulations indicated significant differences in pH levels among the different formulations. The F-statistic of 41.7018 was much higher than the critical value of 2.0595, and the P-value was below the 0.05 significance threshold. These findings demonstrated that the variations in pH between the various NADES formulations were statistically significant. The large F-statistic suggested that the differences in pH between groups were much greater than the variations within each group. Thus, the specific

composition and ratios of components in each NADES formulation significantly impacted the pH levels. The observed differences in pH were not due to random variation but were attributable to the distinct formulations of the NADES. This underscores the importance of precise formulation in determining the pH of NADES, which is crucial for their potential applications and functionality. In short, careful consideration of pH in solvent selection, especially in the context of eco-friendly NADES, is a critical aspect of modern scientific endeavours. By maintaining an appropriate pH range, researchers can ensure the stability, safety, and success of chemical processes, while also exploring new and exciting avenues, such as the development of potent antimicrobial agents. The continued research and understanding of the role of pH in solvent behaviour and applications will undoubtedly lead to further innovations and advancements in various scientific disciplines.

3.3. Density of NADES

The density of NADES exhibited notable range, spanning from 1.0286 g/mL (ChCl:Fru (1:1)) to 1.4067 g/mL (ChCl:Fru (1:2)) as shown in Table 4. This observed spectrum was substantiated by various researchers, providing a comprehensive understanding of NADES density (Koh et al., 2023). The density of NADES stands out as a crucial parameter in the design of chemical processes and the formulation of mathematical equations associated with mass transfer. Opting for high-density NADES may not be optimal for mixing purposes; however, it proves advantageous for extraction due to the substantial density contrast required for effective phase separation. All the NADES exhibited a higher density, measuring 1 g/mL, in comparison to pure water, ethanol, methanol, and hexane, which are common organic solvents with densities below 1 g/mL.

According to the review done by Koh et al. (2023), the author established that the density of NADES incorporating sugar as an initial component fell within the broader range of 1.200–1.460 g/mL. Turning to ChCl:Glu (1:1), the observed density was 1.36065 g/mL, falling between the room temperature densities of pure ChCl (1.1856 g/cm³) and pure Glu (1.5345 g/cm³). This alignment with existing data from Hayyan et al. (2013) indicated consistency, as the density of the same NADES composition with a molar ratio of 1:1 reached a maximum of 1.2978 g/cm³ at room temperature. Similarly, for ChCl:Xyl (1:1), the density was observed at 1.35675 g/mL, which was consistent with the data obtained by Florindo et al. (2017). In this case, the density of the same NADES composition with a molar ratio of 1:1 reached a maximum of 1.257 g/mL at room temperature (Florindo et al., 2017). In the case of ChCl:Xyli (1:1), the obtained density of 1.35855 g/mL aligned with the findings of other researchers, who reported densities ranging from 1.185

to 1.2445 g/mL for ChCl:Xyli (Biernacki et al., 2020; López et al., 2020). The minor differences in density were attributed to the presence of water in the NADES. This addition of water led to a decrease in density, attributed to weakened intermolecular forces such as electrostatic, hydrogen bonds, and Van der Waals forces between the HBA and HBD, ultimately resulted in reduced density. For instance, Biernacki et al. (2020) incorporated 1.21 % water, while López et al. (2020) used NADES with three parts of water which resulted in lower density as compared to presently tested NADES which did not contain water. On top of that, the impact of water content in NADES revealed a decreasing density trend as water content increased from 10 % (v/v) to 50 % (v/v). Notably, densities reduced significantly when exceeding 50 % (v/v) water addition. For instance, the density of the NADES composed of Malic acid and Glu decreased from 1.50 g/cm³ to 1.26 g/cm³ as water content increased from 10 % (v/v) to 50 % (v/v) (Mitar et al., 2019; Santana et al., 2019).

In ChCl:Gly (1:4), the density of pure Gly served as a reference (1.26 g/mL), and the addition of ChCl to Gly was noted for reducing the density of NADES due to an increase in free volume (Omar and Sadeghi, 2022). This signified the impact of the composition on the physical properties of DES.

The comparison between ChCl:Fru (1:1) and ChCl:Fru (1:2) revealed a slight difference in density. The finding of ChCl:Fru (1:2) demonstrating a higher density than ChCl:Fru (1:1) was in alignment with observations from Yang (2019). As reported by the authors, the density of Fru-based NADES increased with an escalation in the molar ratio of Fru to ChCl. This dependency on molar ratio underscored the role of intermolecular interactions, particularly hydrogen bonding, between the components, as stronger or more prevalent interactions significantly impacted molecular arrangement and contributed to density variations. ChCl:Fru (1:2) exhibited stronger intermolecular forces compared to ChCl:Fru (1:1). This observation was further supported by FT-IR analysis, indicating that ChCl:Fru (1:2) had a lower transmission percentage (%T) 3273.24 cm⁻¹ than ChCl:Fru (1:1) at 3276.96 cm⁻¹. The specific -OH peak in the FT-IR spectrum, for instance, provided insights into the strength of intermolecular interactions. This distinction in transmission percentage implied a higher degree of absorption in ChCl:Fru (1:2), indicative of increased intermolecular forces and emphasized the significance of molar ratio in influencing the molecular dynamics and density of the NADES.

Lastly, the ANOVA analysis revealed significant differences in the densities of the various NADES formulations. With an F-statistic of 27.4354, which was substantially higher than the critical value of 4.38737, and a P-value of 0.00046, which was well below the 0.05 significance threshold, the null hypothesis that all NADES formulations have the same density was confidently rejected. This indicated that the observed variations in density were not due to random chance. Therefore, it was evident that the composition and the specific ratios of the components in each NADES formulation exerted significant impact on the densities.

3.4. Polarity of NADES

The polarity of NADES was determined through absorption spectroscopy with the aid of solvatochromic dye Nile red. Nile red dissolved rapidly in ethanol, producing a reddish-pink solution (Farooq et al., 2020). It was observed that the Nile red solution required extensive mechanical stirring for incorporation into the highly viscous NADES. Excess ethanol was then evaporated in the presence of nitrogen gas to prevent interference. The polarity of NADES were calculated from the maximum absorption and presented in E_{NR} (Eq. (2)).

The effectiveness of NADES in extracting substances is closely related to their polarity and miscibility. When NADES exerts a polarity closest to that of the target substances, it proves to be the most suitable choice for the extraction process. This correlation leads to higher yields in the extraction process, as highlighted by Skarpalezos and Detsi (2019).

Table 4
Density and polarity of NADES.

NADES	Density (g/mL)	E _{NR} (kcal mol ⁻¹)
ChCl:Glu (1:1)	1.36065 ± 0.037	49.1677
ChCl:Xyl (1:1)	1.35675 ± 0.004	49.4654
ChCl:Fru (1:1)	1.22860 ± 0.010	49.6372
ChCl:Fru (1:2)	1.40670 ± 0.007	49.3800
ChCl:Gly (1:4)	1.22985 ± 0.004	49.6045
ChCl:Xyli (1:1)	1.35675 ± 0.031	50.0280
ANOVA (Source of Variation)		
Within group		
SS	0.00249	0.0009
df	6	6
MS	0.00041	0.0002
Between group		
SS	0.05683	0.8452
df	5	5
MS	0.01137	0.1690
F	27.4354	1097.0569
P-value	0.00046	8.55E-09
F crit	4.38737	4.3874

The data are expressed as mean ± SD based on duplicate values.

Consequently, the polarity of NADES is considered during solvent selection for extraction procedures. Higher polarity NADES will shift the peak of the solvatochromic dye solution to the right resulting in higher wavelength value and lower E_{NR} value. The polarity of the NADES was determined by obtaining the λ_{max} of solvatochromic dye relative to the value it exhibited in a reference solvent.

Based on Table 4, the E_{NR} of ChCl:Glu (1:1) was 49.1677 kcal mol⁻¹ which was slightly lower than other tested NADES, hence the polarity of ChCl:Glu (1:1) was slightly higher in comparison to the rest in this research. The polarity reduced as the HBD changed from Glu, Fru, Xyl, Gly and Xyli. The polarity of the tested NADES were close to the polarity of common organic solvent such as methanol (51.89 kcal mol⁻¹), ethanol (51.90 kcal mol⁻¹) and water (48.21 kcal mol⁻¹) (Ruesgas et al., 2017). The E_{NR} value of the NADES were consistent to the reported data ranging from 48.2 kcal mol⁻¹ to 50.1 kcal mol⁻¹ (Dai et al., 2013).

The polarity of Xyli-based NADES was observed to be lowest when compared to Glu-based NADES, mainly attributed to their distinctive structural arrangements. In the case of Glu, the hydroxyl groups align along a linear chain, creating a configuration with more accessible and uniformly distributed HBD sites. Conversely, Xyli, classified as a sugar alcohol with five carbon atoms, shares similarities with Glu in its potential to exist in both linear and cyclic forms. However, a key distinction lies in the symmetric arrangement of the OH groups in Xyli. This structural disparity significantly contributed to the observed difference in polarity between Glu and Xyli-based DES, emphasizing the pivotal role of molecular arrangement in influencing the overall polarity of these compounds. Gly-based NADES generally exhibited lower polarity compared to sugar-based NADES, primarily due to the disparity in the number of OH groups. Among all the HBD, Gly has the lowest OH group count, with only 3, while sugar-based HBD have a higher OH group count, ranging from 5 to 6. This discrepancy in OH group quantity served as a determining factor in the observed differences in polarity between Gly-based and sugar-based NADES. Similar reasoning applied to Xyl-based NADES, where it demonstrated the lowest polarity among sugar-based NADES. This was primarily attributed to the comparatively lower quantity of OH groups in Xyl compared to other sugar-based HBD. Xyl contains four OH groups, while both Fru and Glu have five OH groups each. Consequently, Xyl exhibited lower polarity when compared to Glu and Fru due to its reduced OH group content. According to Mulia et al. (2017), the author mentioned that the addition of OH groups present in the forming component resulted in higher polarity value as the electronegativity of the oxygen present in the OH group increased the polarity of the OH group.

It is worth noting that the Fru-based NADES had a slightly lower polarity compared to the Glu-based NADES even though they have the same number of OH group as well as similar amount of carbon atoms. This was primarily due to the difference of structural arrangement, which reduced the number of potential hydrogen bonding sites. Both Glu and Fru are highly polar due to their multiple hydroxyl groups, but Glu is generally considered slightly more polar than Fru. Glu is an aldohexose with an aldehyde group and predominantly exists in a cyclic six-membered ring form (pyranose) in solution. This six-membered ring structure of Glu allowed more extended and linear arrangement of hydroxyl groups, potentially facilitated more hydrogen bonding interactions. In contrast, Fru is a ketohexose with a ketone group and commonly forms a five-membered ring (furanose). This five-membered ring structure of Fru created a more compact structure, which slightly reduced the accessibility of its hydroxyl groups for hydrogen bonding compared to Glu. The ketone group in fructose consists of a carbon-oxygen double bond with the oxygen atom being less electronegative compared to the oxygen in the aldehyde group of Glu. This reduced the ability of the ketone group in Fru to donate hydrogen bonds, making Fru less capable of forming hydrogen bonds with other components compared to Glu. Additionally, the specific arrangement of hydroxyl groups in Fru, coupled with potential steric hindrance effects, further

contributed to its lower accessibility in forming hydrogen bonds. Therefore, the structural differences between Glu and Fru significantly influenced their respective abilities to interact through hydrogen bonding in biological and chemical contexts. Consequently, the aldehyde group in Glu, along with its molecular structure, rendered Glu-based NADES more polar than Fru-based NADES.

The ANOVA results for polarity revealed significant differences among the various NADES formulations. The analysis showed a high F-statistic of 1097.0569, which was much greater than the critical value of 4.3874, and a P-value below the 0.05 threshold for statistical significance. These findings indicated that the differences in polarity among the NADES formulations were highly significant and unlikely to be due to random variation. The large F-statistic suggested that the variation in polarity between different groups was far greater than the variation within each group. Therefore, it was evident that the composition and ratios of the components in each NADES formulation had a substantial impact on their polarity, highlighting the importance of specific formulations in determining their physical properties.

3.5. FTIR analysis

FT-IR measurements were conducted to gain insight into the intermolecular interactions within NADES. Figure S1 depicts the FT-IR spectra of the NADES prepared using the previously mentioned method. Intriguingly, all the NADES exhibited similar spectral trend. NADES with varying Glu molar ratios (1:1, 1:2, and 2:1) exhibited distinct and intense -OH vibrational stretching absorption peaks at 3272.59 cm⁻¹, 3271.50 cm⁻¹ and 3271.95 cm⁻¹ respectively. Similar absorption patterns were observed in NADES containing Xyl and Gal at molar ratios of 1:1, 1:2, and 2:1, ranging from 3256.47 cm⁻¹ to 3272.28 cm⁻¹ and 3271.17 cm⁻¹ to 3273.45 cm⁻¹, affirming the presence of -OH stretching in the eutectic mixture (Dai and Row, 2019; Delgado et al., 2018; Silva et al., 2020; Zain et al., 2021). Furthermore, -OH vibrational stretching was observed in NADES composed of ChCl:Fru (1:1), ChCl:Fru (1:2), ChCl:Gly (1:4), and ChCl:Xyli (1:1) at 3276.96 cm⁻¹ and 3272.24 cm⁻¹, 3305.31 cm⁻¹, and 3305.30 cm⁻¹, respectively. There was also moderate band presence in all the NADES at the wave number from 2878.61 cm⁻¹ to 2930.34 cm⁻¹ which were attributed by the sp³ C-H stretching vibrations. The C-H stretching was preserved from the carbon chains of the HBD used in the preparation of NADES (Zain et al., 2021).

The presence of a peak in the range of 1643.60 cm⁻¹ to 1653.28 cm⁻¹ indicated the presence of the C=O group, a characteristic peak associated with five and six-membered rings. Similar peak was observed by Zain et al. (2021) which further justified current observation. All sugar HBD including Glu, Fru, Xyl, and Gal contained carbonyl groups. Specifically, Glu, Xyl, and Gal featured aldehyde groups (C=O) at the end of their molecules, classifying them as aldose sugars. Fru, another monosaccharide, also possessed a carbonyl group, but in this case, it was part of a ketone functional group. The ketone group (C=O) in fructose was situated within the structure, distinguishing it from aldose sugars. Notably, NADES composed of ChCl:Gly (1:4) and ChCl:Xyli (1:1) lacked this characteristic peak due to the chemical structure of their respective HBD, namely Gly and Xyli. Gly, distinct from sugars such as Glu or Fru, lacked a carbonyl group (C=O) and consisted of a propane backbone with each carbon atom carrying a OH group. Similarly, Xyli, although not having a carbonyl group like monosaccharides, was characterized by five OH groups attached to its carbon atoms. The absence of a carbonyl group aligned with the features of sugar alcohols, which possessed a distinct chemical structure compared to sugars.

The C-H and CH₂ bending vibrations occurred at a wavenumber of 1500–1100 cm⁻¹ which the scissoring vibration occurred at 1450–1420 cm⁻¹ and the wagging vibration occurred at wavenumber of 1320–1300 cm⁻¹. The twisting vibration was found at 1300–1200 cm⁻¹. From Figure S1, the observed peak ranges from 1477.10 cm⁻¹ to 1478.47

cm^{-1} was due to the C-H and CH_2 bending. This peak was attributed to the bending of CH_2 in the alkyl group, as well as the bending and deformation vibrations of the methyl groups within the trimethyl ammonium $(\text{CH}_3)_3 \text{N}^+$ ion presented in the ChCl molecule (Dai and Row, 2019; Delgado et al., 2018; Zain et al., 2021).

In general, at the fingerprint region of the IR spectra of all the tested NADES in the range of 950 cm^{-1} to 1350 cm^{-1} were attributed to the C-O, C-C-O, C-O-C, C=C and C-N stretching and C-O-H bending vibrations. Similar band was observed by Dai & Row (2019) and Zain et al. (2021) on all the similar NADES. It is worth noting that, in ChCl:Gly, the band observed at 1033 cm^{-1} was associated with the C-O linkage between C_1 and C_3 of Gly. Simultaneously, the band at 1108 cm^{-1} was attributed to the C_2 C-O stretching vibration (Zain et al., 2021).

It is also worth noting that the intensity of OH-stretching, sp^3 C-H stretching, H-O-H scissoring vibration, six-membered ring, C-O, C-C-O, C-O-C, C=C and C-N stretching and C-O-H bending vibrations increased as the molarity of the sugar-based NADES changed from 2:1, 1:1 and 1:2. Hence, this indicated additional bonds were formed with increased amount of sugar presented in the NADES. This finding was obvious because sugar molecule were responsible for those bonds (Dai and Row, 2019; Zain et al., 2021). On the other hand, the intensity of C-H bending increased as the molarity of sugar-based NADES changed from 1:2, 1:1 and 2:1. According to Zain et al. (2021), ChCl was responsible for the sharp intense peak at 1483 cm^{-1} which was contributed by the C-H bending and the deformation vibration of the three methyl groups attached to the ammonium ion. Thus, the intensity of the peak increased as the amount of ChCl presented in the NADES increased.

In conclusion, FT-IR measurements were conducted to investigate intermolecular interactions in NADES. The spectra obtained from FT-IR,

exhibited a consistent trend across all NADES. Various vibrational peaks were identified, providing insights into the composition of NADES with different sugar molar ratios and distinct components. The study also explored the impact of molarity on peak intensities, revealing potential bonding variations.

3.6. Rheology assessment of NADES

All the formulated NADES exhibited wide range of high viscosities, spanning from $17117 \text{ mPa}\cdot\text{s}$ to $275000 \text{ mPa}\cdot\text{s}$. These values significantly surpassed the viscosity levels observed in common organic solvents such as water ($1.0016 \text{ mPa}\cdot\text{s}$), ethanol ($1.095 \text{ mPa}\cdot\text{s}$), methanol ($0.5600 \text{ mPa}\cdot\text{s}$), and acetone ($0.316 \text{ mPa}\cdot\text{s}$). The elevated viscosity within NADES was attributed to the extensive hydrogen bonding between its starting components. While this characteristic enhanced the overall viscosity, it introduced challenges by slowing down mass transfer, extraction, and dissolving processes, consequently increased the costs associated with pumping. Nevertheless, the viscosity of NADES was not a fixed parameter and can be manipulated. This can be achieved through strategic additions of water and adjustments in temperature, as water weakened the hydrogen bonds among the components, thereby reduced the overall viscosity.

Fig. 1 (a) reveals that the viscosity of NADES experienced an increase as the HBD changed from Gly, Xyli, Xyl, Fru, to Glu. Gly, characterized by a shorter carbon chain, exhibited the lowest viscosity. As outlined in Cao et al. (2022), the trend indicated that the viscosity of NADES tended to increase as the length of the carbon chain became longer. Gly stood out from other HBD due to its distinctive three-carbon atom chain, contrasting with the longer five-carbon chains found in Xyl and Xyli, as well as the six-carbon chains presented in Fru and Glu. This nuanced

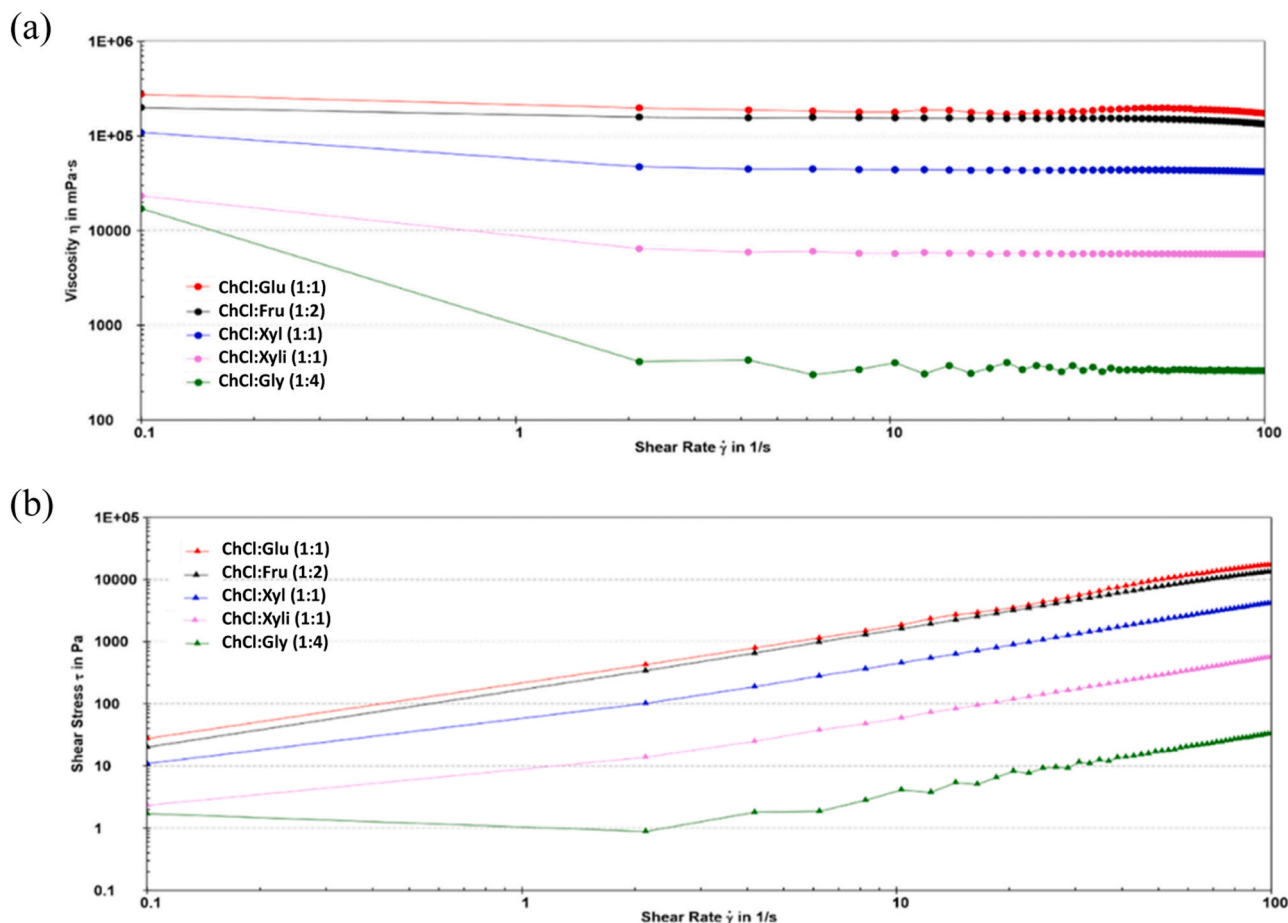


Fig. 1. Viscosity profiles of NADES across shear rates (a) and shear stress dynamics across shear rates (b).

relationship between carbon chain length and viscosity underscored the distinct properties of these HBD. Moreover, the molecules of the HBD exhibited a higher tendency to become entangled as the length of their carbon chains increased. This entanglement phenomenon impeded the free movement of HBD, leading to a noticeable elevation in the viscosities of NADES.

The freezing point emerged as a critical factor influencing viscosity, as indicated by findings presented by Hayyan et al. (2012). It is established that a lower freezing point correlated with lower viscosity. A concrete illustration of this relationship was observed by the authors. For instance, ChCl:Fru (3:1) stood out with the lowest freezing point at 10°C, and correspondingly, it exhibited the lowest viscosity among all the NADES tested. This aligned seamlessly with our observations, particularly evident in Gly, which obtained the lowest freezing point. Examining the freezing points of the HBD provided further insight. The freezing point ascended in the order of Gly (18.2°C), Xyli (90–92°C), Fru (103°C), Xyl (140°C) and reached the highest point with Glu (146°C). Notably, this freezing point trend was mirrored in the viscosity of the NADES. It is worth noting that even though Xyl exhibited a higher freezing point compared to Fru, yet its viscosity was lower than that of Fru. This discrepancy was a consequence of the distinct carbon chain lengths in the two compounds. Viscosity is a multifactorial property influenced by various factors, including the carbon chain length and the freezing point of the components. The interplay of these factors contributed to the nuanced variations in viscosity observed in Xyl, highlighting the complex nature of the relationship between freezing points, viscosities, and molecular structures in these substances.

Analyzing the change of viscosity over shear rate provided valuable insights into the rheological behavior of various NADES. For ChCl:Glu (1:1) and ChCl:Fru (1:2) NADES, the viscosity remained consistently constant despite an increase in shear rate. This behavior indicated that both ChCl:Glu (1:1) and ChCl:Fru (1:2) NADES exhibited Newtonian fluid characteristics. Conversely, the viscosity of ChCl:Xyl (1:1) and ChCl:Xyli (1:1) NADES showed a slight decrease as the shear rate increased from 0.1 s⁻¹ to 2.14 s⁻¹, with viscosity dropping from 109000 mPa.s to 47252 mPa.s for ChCl:Xyl (1:1) and from 23288 mPa.s to 6455 mPa.s for ChCl:Xyli (1:1). However, beyond a shear rate of 2.4 s⁻¹ up to 100 s⁻¹, the viscosity remained constant for both ChCl:Xyl (1:1) and ChCl:Xyli (1:1) NADES. Of particular note was the significant decrease in viscosity observed for ChCl:Gly NADES, plummeting from 17117 mPa.s to 413.99 mPa.s as the shear rate increased from 0.1 s⁻¹ to 2.4 s⁻¹. Notably, the viscosity maintained constancy as the shear rate continued to rise beyond 2.4 s⁻¹ up to 100 s⁻¹.

For ChCl:Xyl (1:1) and ChCl:Xyli (1:1) NADES, the viscosity exhibited a slight drop and then remained constant, showing some resemblance to Bingham plastic behavior. However, analyzing the shear stress versus shear rate graph as shown as Fig. 1 (b) indicated that the shear stress for both NADES increased linearly with the shear rate. This linear relationship between shear stress and shear rate signified that both ChCl:Xyl (1:1) and ChCl:Xyli (1:1) NADES behaved as Newtonian fluids. The minor reduction in viscosity was attributed to structural reorganization, where the internal arrangement of the NADES underwent a slight reorganization or alignment as the shear rate increased. This structural adjustment led to a decrease in resistance to flow, resulting in a subsequent reduction in viscosity.

In contrast, ChCl:Gly (1:4) exhibited Bingham plastic behavior. As shown Fig. 1 (b), the shear stress remained constant as the shear rate increased from 0.1 s⁻¹ to 2.14 s⁻¹. Beyond this point, as the shear rate increased further, the shear stress rose linearly. The constant shear stress within the initial shear rate range corresponded to the yield stress which was the minimum stress required to initiate flow in the material (Zengeni, 2016). Below this critical stress, ChCl:Gly (1:4) behaved like a solid, resisting deformation. Beyond the yield stress, the graph transitions to a linear region, known as the yield point. In this linear segment, the relationship between shear stress and shear rate became proportional, resembling Newtonian behavior. The slope of this linear portion

represented the viscosity of the material in the Newtonian regime. The distinctive features of the Bingham plastic graph underscored its dual nature whereby the initial solid-like behavior below the yield stress, followed by a more fluid-like response above the yield stress (Zengeni, 2016).

In summary, the study highlights the diverse rheological behaviors of NADES. The significant viscosity range, influenced by hydrogen bonding, freezing points, and carbon chain lengths, offers insights into their complex fluid dynamics. The adaptability of NADES viscosity to external factors underscores their potential for tailored applications in various fields.

3.7. Phenolic compounds measurement

The TPC in the OPL was determined via the FC reagent, which is used to determine the total phenolic and polyphenol content by determining the reductive capacity of the antioxidant (Ahmad et al., 2018; Noreen et al., 2017; Lamuela-Raventós, 2018). Alternative methodologies, such as the 2,2-diphenyl-1-picrylhydrazyl (DPPH) assay, liquid chromatography, and gas chromatography, have been employed by previous researchers for the quantification of phenolic compounds. In current investigation, FC assay was chosen as interest lied in the determination of the TPC, which is a parameter that provides insight into the phenolic composition of the samples. While the DPPH assay is adept at assessing the antioxidative scavenging activity against the stable DPPH radical, it does not capture the entirety of the phenolic content. Furthermore, when considering the application of liquid chromatography and gas chromatography, both methodologies entail intricate sample preparation processes, which inherently demand substantial investment of time and cost. Method development for chromatographic technique is characterized by its challenges and necessitates extensive preliminary experimentation to ascertain optimal parameters. Additionally, the procurement, operation, and maintenance of equipment associated with these chromatographic methods are accompanied by notable financial and technical complexities. Given these considerations, a comprehensive evaluation led to the utilization of the FC assay. This decision aligned harmoniously with the research objectives and practical constraints, ultimately contributed to the scientific rigor and feasibility of the analytical approach.

Phenolic and polyphenol compounds, secondary metabolites within plants, exhibit detectability through UV-Vis spectrophotometry by employing FC reagents. The constitution of the FC reagent encompasses sodium tungstate and sodium molybdate within an acidic medium. Upon interaction with phenolic compounds, the reagent, initially manifesting a yellow hue, undergoes reduction, which facilitated by the phenolic compounds present within the OPLE, thereby transitioning to a distinct blue coloration. The ensuing alteration in colour and its corresponding intensity following this reaction is subsequently analysed through UV-Vis spectrophotometric analysis, performed at a wavelength of 765 nm (Lizcano et al., 2019). In this analysis, the standard for measuring the TPC is gallic acid. TPC in the OPL was determined through the standard calibration graph with a correlation coefficient (R^2) of 0.9992 by the standard calibration curve equation obtained at $y = 0.8808x + 0.0056$. Thus, the data fitted well in the standard calibration graph as $R^2 > 0.995$. The equation obtained was then used in the following concentration calculation for TPC in OPL-NADES extracts.

As mentioned previously, NADES has close polarity compared with common organic solvents as well as water. Therefore, to ensure the formation of a homogeneous liquid, 0.1 mL of the NADES OPL extract was dissolved in 7 mL of ultrapure water. The homogeneity of the NADES extract was pivotal in obtaining accurate results for the TPC. Subsequently, the dissolved NADES OPL extract was combined with 0.5 mL of a 10 % FC reagent solution. The prepared reaction mixture was carefully stored in darkness for a duration of 10 minutes. This incubation period facilitated the interaction between the phenolic

compounds and the FC reagent, initiated the colorimetric response. To establish an alkaline setting, 1.5 mL of a 7.5 % sodium carbonate solution was introduced into the mixture. The addition of sodium carbonate was pivotal in creating an alkaline environment because the most effective interaction between the phenolic compounds and the FC reagents occurred in alkaline conditions. Within this alkaline environment, the protons contained in phenolic compounds underwent dissociation, resulting in the creation of phenolic ions. As the reaction continued, a blue shade became apparent. This alteration in colour signified an increased concentration of generated phenolic ions. Higher levels of phenolic compounds contributed to a more notable reduction of heteropoly acid by these phenolic ions, thereby amplified the intensity of the resulting blue shade. Subsequently, the reaction mixture was allowed to incubate at room temperature for a duration of 2 hours. This extended incubation period was designed to optimize the reaction kinetics, ensuring the comprehensive evolution of the colorimetric response. The final step involved the measurement of absorbance. Using a UV-Vis spectrophotometer, the absorbance of the reaction mixture was quantified at a specific wavelength of 765 nm. This wavelength corresponded to the optimal absorbance of the colorimetric product generated as a result of the reaction between the phenolic compounds and the FC reagent. By meticulously adhering to this standardized protocol, the TPC in the NADES OPL extract was precisely measured, thereby contributing to the comprehensive characterization of the botanical constituents presented within the NADES extract.

3.8. Carotenoids measurement

The solubility behavior of the NADES extract in various solvents was assessed, revealing its insolubility in hexane and acetone. This posed a challenge for the analysis of carotenoids due to immiscibility with these typically effective carotenoid-extraction solvents. To address this, solid phase extraction using a Bond Elut Plexa cartridge (200 mg, 3 mL) was employed. The preparation of the cartridge involved a conditioning process in which methanol and subsequently distilled water were introduced to activate the adsorbent. The highly viscous nature of the NADES extract necessitated its dilution with 5 mL of distilled water and thorough vortexing to achieve homogeneity. The diluted NADES extract was then loaded into the solid phase extraction cartridge, with subsequent outflows discarded while retaining the extract within the cartridge. To ensure comprehensive carotene extraction, the cartridge underwent two rounds of rinsing with acetone, followed by a final rinse with hexane to ensure no residual carotene remained. The outflow exhibited a two-phase separation: an upper phase containing chlorophyll and carotene, and a lower phase comprising water and NADES. The lower phase was removed using a 1 mL syringe rinsed with acetone, while the upper phase, denoted as the sample, was transferred to a 5 mL measuring cylinder, with the volume measured and recorded.

Considering the presence of chlorophyll in the sample, the Ambersep 900OH resin was employed at a ratio of 1 g resin to 10 mL sample solution. The mixture was stirred at 300 rpm and room temperature for 30 minutes. The resin was then removed, replaced with fresh resin, and subjected to an additional 30 minutes of stirring. An alternative approach for chlorophyll removal involves saponification. However, this method was not chosen due to its reliance on heat, which exerts detrimental effects on carotenoid compounds due to their susceptibility to thermal degradation. Saponification processes are typically protracted, often entailing prolonged reaction times, and necessitate the use of potent alkaline solutions that result in by-product formation and waste generation. Conversely, the selection of Ambersep 900 OH circumvents these drawbacks. This resin can be regenerated and its application avoids the need for heating, thereby offering a more controlled and efficient means of chlorophyll removal. Subsequently, the absorbance was measured via UV-Vis spectrophotometry at a wavelength of 450 nm, using acetone as the blank. This comprehensive procedure facilitated precise determination of carotenoid content within the

NADES extract.

The TCC of the standard calibration solution, spanning concentrations of 0.1, 0.5, 1.5, 3.0, 6.0, and 9.0 µg/mL, was obtained in terms of total carotenes equivalent, expressed in µg/g of the extract. The calibration data were subjected to UV-Vis spectrophotometry analysis at 450 nm, culminating in the derivation of the standard calibration equation from empirical observations, represented as $y = 0.2603x + 0.0098$, with a R^2 of 0.9997, surpassing the threshold of 0.995. This high R^2 value validated the adherence of the data to the established standard calibration graph. The ensuing equation was subsequently utilized for TCC computations, facilitating accurate measurement of the carotenoid content in the extract.

The validity of the findings was underpinned by key parameters such as the molar extinction coefficient (ϵ) and the R^2 . In this study, a ϵ of 0.2603 was obtained, signifying the relationship between concentration and absorbance. The high R^2 value of 0.9997 achieved in this work underscored the strength of association between the variables. Comparatively, with reference to Hagos et al. (2022), a ϵ of 0.2557, aligning with a R^2 of 0.99995 were reported. These parameters collectively contributed to the robustness and reliability of current findings, supporting the credibility of the work.

3.9. Performance of NADES on TPC and TCC yield

In this scientific exploration, the TPC obtained using five different NADES were investigated, as illustrated in Fig. 2. ChCl:Xyl (1:1) exhibited the highest TPC (20.23825 mg/g), whereas the TPC was the lowest with ChCl:Glu (1:1) (11.43785 mg/g). Phenolic compounds encompassed one or more OH, recognized as the polar segment, directly bonded to an aromatic ring, forming the non-polar segment (Galanakis et al., 2013). Due to their polar nature, phenolic compounds exhibited enhanced solubility in polar solvents, consequently yielding a higher TPC compared to non-polar solvents. In Section 3.4, it was established that all the tested NADES exhibited polar natures, and their polarity can be arranged in the following sequence: ChCl:Xyl (1:1) (50.0280 kcal mol⁻¹) < ChCl: Gly (1:4) (49.6045 kcal mol⁻¹) < ChCl: Xyl (1:1) (49.4654 kcal mol⁻¹) < ChCl: Fru (1:2) (49.3800 kcal mol⁻¹) < ChCl: Glu (1:1) (49.1677 kcal mol⁻¹). This sequence denotes a progression of increasing polarity with a decrease in the E_{NR} value. Intriguingly, ChCl: Xyl (1:1), despite manifesting intermediate polarity among the polar NADES, demonstrated a notably superior TPC yield. The TPC yield exhibited a downward trend as polarity decreased and further diminished with an increase in polarity as shown in Fig. 3. Notably, ChCl:Fru (1:2) and ChCl:Glu (1:1), characterized by the highest polarity among all tested NADES, acquired the lowest TPC yield. This observation suggests that the optimal polarity for extracting phenolic compounds from OPL was attained with ChCl:Xyl (1:1), and polarities deviating higher or lower resulted in a reduction of TPC.

Additionally, it is noteworthy that ChCl:Xyl (1:1) also demonstrated an intermediate viscosity compared to all the chosen NADES for the extraction process as discussed in Section 3.6. This observation suggested that the viscosity of ChCl:Xyl (1:1) with 10 wt% of water was conducive to the extraction of phenolic compounds from OPL. The intermediate viscosity implied an optimal balance for facilitating the extraction process, signifying favourable conditions for the efficient retrieval of phenolic compounds from OPL using ChCl:Xyl (1:1). This aspect is crucial as it relates to mass transfer during extraction, where an optimal viscosity ensured a suitable balance between solute diffusion and fluid flow.

In addition to considering the polarity and viscosity of NADES, the pH of these solvents also emerges as one of the factors influencing the efficiency of extracting phenolic compounds from OPL. Despite ChCl: Glu (1:1) and ChCl:Fru (1:2) exhibiting higher polarity than ChCl:Xyl (1:1), they paradoxically yielded lower TPC as depicted in Fig. 2 and Fig. 3. The disparity in TPC between ChCl:Glu (1:1) and ChCl:Fru (1:2)

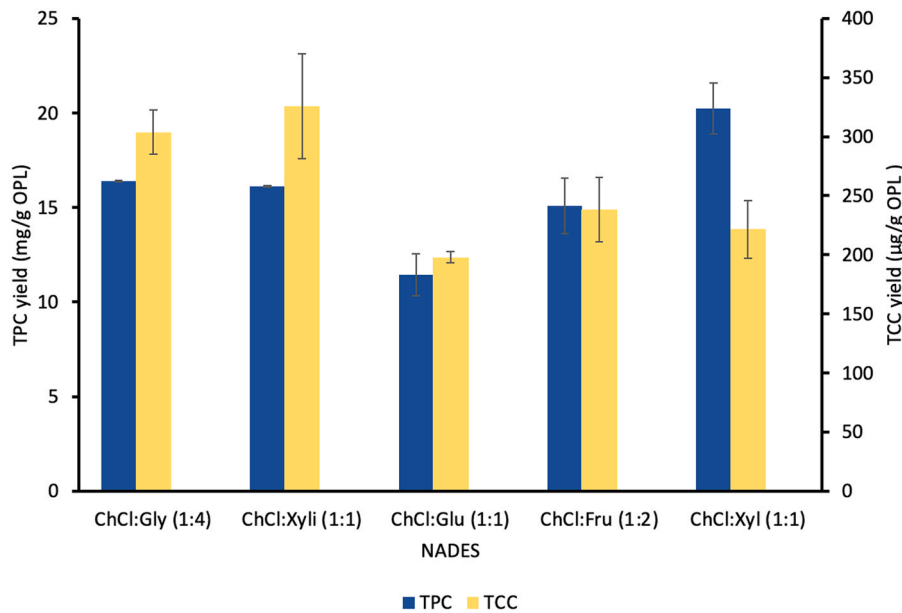


Fig. 2. Performance of different NADES on TPC and TCC yield.

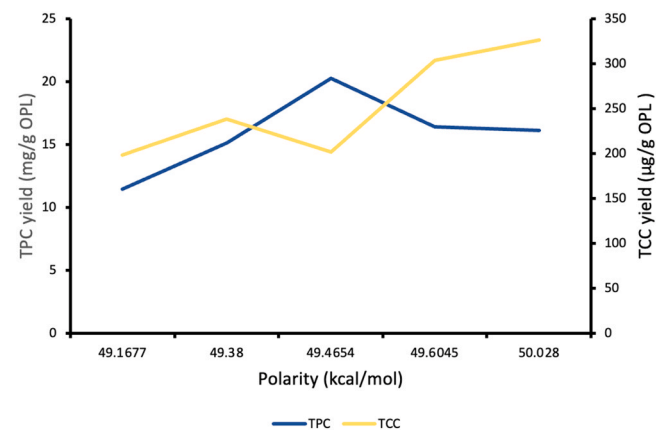


Fig. 3. TPC and TCC yields over polarity.

despite having similar viscosity, molecular weight, and polarity, primarily stemmed from the difference in pH between these NADES. ChCl:Fru (1:2) maintained a slightly acidic pH of 5.82, while ChCl:Glu (1:1) featured a neutral pH of 6.72. The reason for specifically comparing Glu and Fru was due to their similar physicochemical properties, with the exception of pH. This approach minimized the influence of other variables, allowing for a more precise evaluation of the impact of pH on extraction efficiency. Phenolic compounds was found sensitive to pH according to the findings of Ruenroengklin et al. (2008), who observed that phenolic compounds were stable within the pH range of 1–5, attaining peak TPC at pH 4. Similar observation was obtained by Várfalvyová et al. (2023) when extracting polyphenolic compounds from different sources using NADES. The author observed that higher polyphenol was extracted at acidic to slightly acidic condition as compared to neutral condition. The highest polyphenol yield was obtained at pH 3 and it reduced as the pH increased till 7. Deviations from this pH range, either towards high or low values, were noted to potentially induced instability and decomposition of phenolic compounds. Furthermore, according to Engy (2019), the extracted TPC reduced as the pH changed from acidic to neutral. The TPC content reduced from 46.1 mg TPC/ g to 25.5 mg TPC/ g from a peanut meal as the pH rose from pH 1 to pH 6. It can be concluded that pH impacted the stability

and TPC in extract. As a consequence, ChCl:Fru (1:2) demonstrated greater extraction of phenolic compounds in comparison to ChCl:Glu (1:1) at neutral pH conditions. This pH variation highlighted the significant impact of the acidity or neutrality of the NADES on the extraction efficiency of phenolic compounds from OPL, underscoring the intricate interplay between solvent characteristics and extraction outcomes.

The ANOVA results for total phenolic extraction using different NADES formulations in Table 5 revealed significant differences in the extraction efficiency. The F-statistic of 19.1882 was much higher than the critical value of 5.1922, and the P-value of 0.0031 was well below the 0.05 threshold, indicating that the variations in TPC between the different NADES formulations were statistically significant. This suggested that the differences in phenolic extraction are not due to random variation but are attributable to the specific compositions and ratios of the NADES formulations. The large F-statistic underscored that the variation in phenolic content between groups was far greater than within each group, highlighting the importance of precise formulation in enhancing the efficiency of phenolic extraction. This is crucial for applications where phenolic compounds are valuable, such as in the food,

Table 5
Performance of NADES on phytonutrients extraction.

Term	TPC (mg/g OPL)	TCC (µg/g OPL)
NADES		
ChCl:Gly (1:4)	16.41 ± 0.04	303.62 ± 18.74
ChCl:Xyli (1:1)	16.12 ± 0.04	325.94 ± 44.39
ChCl:Glu (1:1)	11.44 ± 1.13	197.93 ± 4.55
ChCl:Fru (1:2)	15.08 ± 1.46	237.99 ± 27.17
ChCl:Xyl (1:1)	20.24 ± 1.33	221.48 ± 24.18
ANOVA (Source of Variation)		
Within group		
SS	5.1728	3665.59
df	5	5
MS	1.0346	733.1189
Between group		
SS	79.4047	24075.36
df	4	4
MS	19.8512	6018.8409
F	19.1882	8.2099
P-value	0.0031	0.0201
F crit	5.1922	5.1922

The data are expressed as mean ± SD based on duplicate values.

pharmaceutical, and cosmetic industries.

This study also examined the extraction efficiency of NADES on carotenoids from fresh OPL. The highest TCC was identified in ChCl:Xyli (1:1), reaching 325.94 µg/g OPL followed by ChCl:Gly (1:4) at 303.62 µg/g OPL. In contrast, the lowest TCC was observed in ChCl:Glu (1:1), with a value of 197.93 µg/g OPL. The significant variation in TCC between these NADES was attributed primarily to differences in polarity. The role of polarity is a significant determinant in the extraction of carotenoids. Carotenoids, commonly acknowledged as non-polar compounds, owe this classification to the hydrophobic characteristics inherent in their molecular structure. The defining feature contributing to their nonpolar attributes is the presence of extensive conjugated hydrocarbon chains, coupled with a polyene chain characterized by alternating single and double bonds. Prominent in fruits and vegetables, carotenoids such beta-carotene and lycopene exhibit limited solubility in water but readily dissolve in non-polar solvents. As highlighted in Section 3.4, NADES formulations, such as ChCl:Xyli (1:1) ($E_{NR} = 50.0280 \text{ kcal mol}^{-1}$) and ChCl:Gly (1:4) ($E_{NR} = 49.6045 \text{ kcal mol}^{-1}$), demonstrated the lowest polarity, surpassing ChCl:Glu (1:1) ($E_{NR} = 49.1677 \text{ kcal mol}^{-1}$). The polarity of NADES was linked to their extractability and miscibility, with formulations closely aligned in polarity with the target substances emerging as optimal choices during the extraction process (Koh et al., 2023). This correlation resulted in enhanced extraction effectiveness, leading to higher yield. In this specific context of carotenoid extraction, NADES with lower polarity was proven favourable as illustrated in Fig. 3. This characteristic facilitated higher extraction of carotenoids from the OPL in comparison to NADES with higher polarity, as evidenced by the variance in carotenoid yield observed between ChCl:Glu (1:1) and ChCl:Xyli (1:1) as well as ChCl:Gly (1:4).

In addition to the influence of polarity, the role of viscosity is a significant determinant in the extraction of carotenoids. In Section 3.6, it was noted that ChCl:Gly (1:4) exhibited the lowest viscosity at 17117 mPa.s, followed by ChCl:Xyli (1:1) at 23299 mPa.s, while ChCl:Glu (1:1) displayed the highest viscosity at 275000 mPa.s. The elevated viscosity of ChCl:Glu (1:1) NADES hindered mass transfer, extraction, and the dissolving process of carotenoids in the OPL, leading to a reduction in carotenoid extraction yield. This observation aligned with findings from a study by Lazzarini et al. (2022), where NADES composed of menthol and lactic acid exhibited the lowest carotene content in tomato pomace due to its high viscosity. A similar conclusion was drawn by Leite et al. (2021), who reported that the high viscosity of ChCl-based NADES, attributed to a synergic effect, created challenges in the interaction between NADES and carotenoids. Consequently, it did not enhance the ethanolic extraction yield of carotenoids from buriti fruit. Further supporting evidence from Li et al. (2022) revealed that a microemulsion formulated using NADES composed of menthol and octanoic acid, exhibiting the highest viscosity (2.7-fold higher than the least viscous NADES), obtained the lowest carotenoid yield. The carotenoids, including astaxanthin, astaxanthin ester, and lutein, were two to five-fold lower compared to the highest yield obtained by the microemulsion formulated with least viscous NADES.

The pH of NADES is also crucial in determining the overall carotenoid extraction yield. Prior research indicated that a more effective extraction could be attained under acidic conditions, approximately in the pH range of 4.0–5.0. The same was observed in carotenoid yield between the slightly acidic NADES denoted by ChCl:Fru (1:2) and the neutral NADES denoted by ChCl:Glu (1:1). ChCl:Fru (1:2), which was slightly acidic, had higher carotenoid yield compared to ChCl:Glu (1:1) with a neutral pH. The carotenoid yield increased by 16.83 % as the NADES used switched from ChCl:Glu (1:1) to ChCl:Fru (1:2). The observation aligned with the conclusion made by other researchers. For instance, Bell et al. (2016) highlighted that solvents with pH ranging from 3 to 6 enhanced the TCC by 22 %, 27 %, 22 %, and 18 % respectively. Their research indicated that the solvent with a pH of 4 exhibited

the highest TCC, while neutral to slightly basic conditions led to a reduction in TCC by 26 %. The carotenoids displayed sensitivity to the pH levels of the solvents employed. Similar conclusions were drawn by Li et al. (2022), where the carotenoids yield increased as the pH of the microemulsion synthesized by NADES reduced. The carotenoids (astaxanthin, astaxanthin ester, and lutein) obtained were 2–5-fold higher in an acidic microemulsion formulated by NADES (pH 4.5) compared to the lowest yield obtained by the high pH microemulsion formulated by NADES (pH 6). This divergence in pH levels between the ChCl:Fru (1:2) and ChCl:Glu (1:1) formulations was instrumental in influencing the solubility and stability of carotenoids. The slightly acidic nature of ChCl:Fru NADES enhanced the solubility and retention of these carotenoids, contributing to the observed elevation in TCC yield.

The ANOVA results for total carotenoid extraction using different NADES as shown in Table 5 indicate significant differences between the groups. The calculated F-value of 8.2099 exceeded the critical F-value of 5.1922, and the P-value of 0.0201 was less than the standard significance level of 0.05. These findings suggested that the variability between groups, attributable to the different NADES, was significantly greater than the variability within groups. Hence, the differences in carotenoid extraction efficiency were more pronounced between the different NADES than the natural variations observed within each NADES group. Consequently, the null hypothesis was rejected and the choice of NADES significantly affected the extraction of total carotenoids was concluded.

The efficiency of NADES in extracting phenolic compounds and carotenoid from fresh OPL was found multifactorial, and a comprehensive assessment involved various parameters rather than relying on a single factor. It is also worth noting that NADES with a lower density is generally easier for separation during centrifugation. A lower density NADES creates a greater density difference between the extract and the OPL fibres. Such difference resulted in faster settling of the denser OPL particles to the bottom of the centrifuge tube under the influence of centrifugal force. The resulted clearer phase separation ensured easier and efficient solid-liquid separation. ChCl:Gly (1:4) yielded the highest volume of extract ($1.645 \pm 0.078 \text{ mL}$) at the lowest density ($1.230 \pm 0.004 \text{ g/mL}$). Additionally, the lower viscosity at 17117 mPa.s enhanced the mobility of the OPL particles, further aiding in their rapid settling. Our research observations suggested a balanced combination of characteristics, including intermediate polarity, viscosity, and a slightly acidic pH, which contributed to the optimal extraction of phenolic compounds from OPL. It emphasized the multifactorial nature of the extraction process and underscored the importance of considering a combination of solvent properties for enhanced efficiency in extracting bioactive compounds from plant materials. For instance, the intermediate polarity of NADES ensures adequate solubility of the target compounds, the low viscosity facilitates proper penetration and mass transfer, and the slightly acidic pH enhance the stability of phenolic compounds. As for carotenoid extraction, the highly viscous ChCl:Glu (1:1) hindered mass transfer, resulting in lower carotenoid yields. Lower polarity in NADES formulations, such as ChCl:Xyli (1:1) and ChCl:Gly (1:4), were proven more effective in extracting carotenoids compared to higher-polarity ChCl:Glu (1:1). Additionally, pH played a crucial role, with slightly acidic conditions in ChCl:Fru (1:2), enhancing carotenoid yields compared to neutral conditions in ChCl:Glu (1:1). The study underscored the importance of considering viscosity, polarity, pH and density in selecting optimal NADES formulations for efficient phenolic compounds and carotenoid extraction. Therefore, it is not merely one property but the interplay of these multiple factors that determines the success of the simultaneous extraction process. The multifactorial approach ensures that the diverse and interactive nature of solvent properties is fully leveraged, leading to more effective and efficient extraction outcomes.

4. Conclusion

In conclusion, current study emphasized the importance of understanding the polarity, pH, viscosity and density of NADES, as these were key factors influencing the extraction yields. NADES also addressed the challenge posed by high moisture content in OPL, which acted as a barrier between bioactive compounds and non-polar extraction solvents. Ultimately, the findings contributed valuable insights into sustainable practices for maximizing the utilization of oil palm biomass, offering a potential avenue for the development of eco-friendly extraction methods in the palm oil industry. This research underscored the need for further exploration and optimization of extraction processes to enhance the efficiency and sustainability of utilizing oil palm waste for valuable bioactive compounds.

CRediT authorship contribution statement

Qi Qi Koh: Writing – original draft, Formal analysis. **Suyin Gan:** Supervision, Resources. **Yin Leng Kua:** Writing – review & editing, Conceptualization. **Yingzhan Zhao:** Investigation. **Zhi Ling Chew:** Investigation. **Harrison Lik Nang Lau:** Supervision, Resources. **Terri Zhuan Ean Lee:** Supervision, Resources. **Khang Wei Tan:** Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fbp.2024.07.023](https://doi.org/10.1016/j.fbp.2024.07.023).

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